SALES PERFORMANCE ANALYSIS DASHBOARD

Final Data Analytics Project

Submitted By:
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Internship – Week 6

Tools Used:
Microsoft Power BI • Git & GitHub

Final Capstone Project

Task 1: Git & GitHub Basics

Objective:
To understand the basics of Git and GitHub and learn how to create a repository, clone it to the local system, add project files, commit changes, and push the changes to GitHub.

Activities Performed:

1. Installed Git on the system.
2. Created/used a GitHub account.
3. Created a new GitHub repository.
4. Cloned the repository to the local system.
5. Added project files to the repository.
6. Created a Git commit.
7. Pushed the changes to GitHub.

Result:
Successfully created and connected a local Git repository with GitHub and pushed the project files to the GitHub repository.

Task 2: Git Workflow Practice

Objective:
To practice the basic Git workflow and understand how to manage project changes using Git and GitHub.

Activities Performed:

1. Used `git clone` to clone the GitHub repository to the local system.
2. Used `git add` to add project files for tracking.
3. Used `git commit` to save changes with meaningful commit messages.
4. Used `git push` to upload local changes to GitHub.
5. Used `git pull` to fetch and update the local repository with changes from GitHub.
6. Created at least **3 meaningful commits** while working on the final project.

Result:
Successfully practiced the basic Git workflow and synchronized the local project with the GitHub repository.

Task 3: Dataset Selection & Data Preparation

Objective:
To select a suitable real-world dataset, inspect its structure and prepare the data for further analysis.

Dataset Selected:
Sales Analytics Dataset

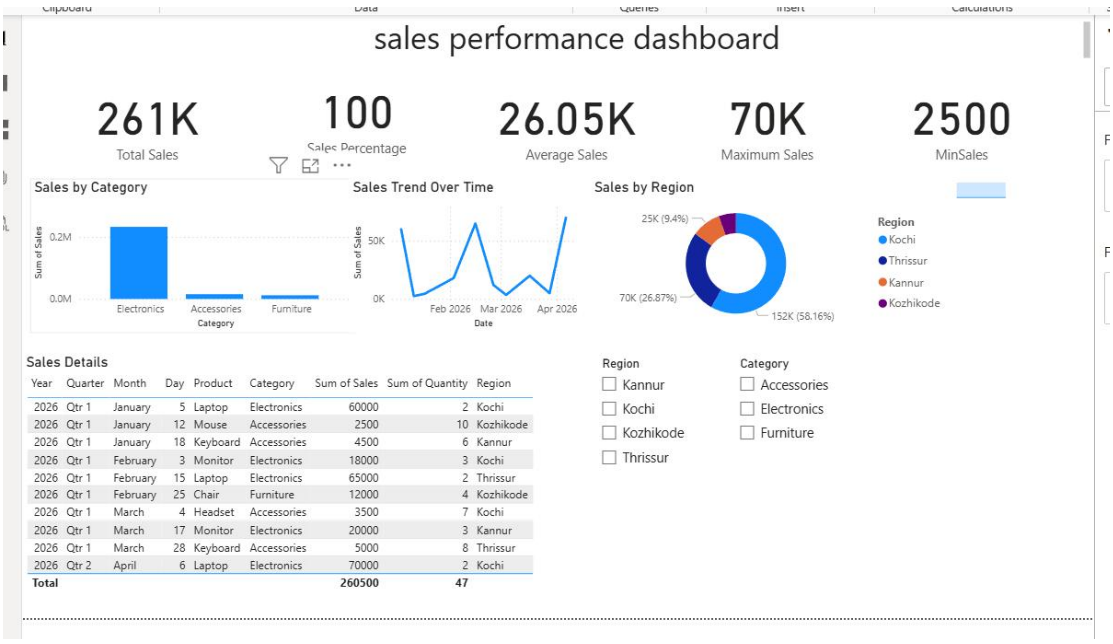
Activities Performed:

1. Selected a real-world **Sales** dataset.
2. Inspected the dataset and reviewed the available records.
3. Identified the columns and their data types.
4. Checked the dataset for missing values.
5. Handled missing values wherever required.
6. Checked for duplicate or incorrect records and removed them where required.
7. Prepared the cleaned dataset for further analysis and visualization.

Dataset Columns:

- Date
- Product
- Category
- Sales
- Quantity
- Region

Result:
The sales dataset was inspected, cleaned, and prepared for **EDA and Power BI visualization**.



Task 4: Exploratory Data Analysis (EDA)

Objective:
To analyze the prepared sales dataset and identify important statistics, patterns, trends, relationships, and outliers.

Activities Performed:

1. Performed descriptive statistical analysis on the dataset.
2. Calculated key measures such as total, average, minimum, and maximum sales.
3. Inspected the data to understand the distribution of sales and quantity.
4. Analyzed the relationship between **Sales** and **Quantity**.
5. Identified important patterns and trends in the sales data.
6. Identified high- and low-performing categories and regions.
7. Checked for unusually high or low sales values (outliers).

Key Findings:

- Total Sales: **₹260,500**
- Average Sales: **₹26,050**
- Maximum Sales: **₹70,000**
- Minimum Sales: **₹2,500**
- **Electronics** is the highest-performing category.
- **Kochi** is the highest-performing region.
- Sales performance varies across the available time periods.

Result:
The exploratory analysis helped identify the major sales patterns, trends, high-performing areas, and important observations in the dataset.

Task 4: Exploratory Data Analysis (EDA)

Objective:
To analyze the prepared sales dataset and identify important statistics, patterns, trends, relationships, and outliers.

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1. Performed descriptive statistical analysis on the dataset.
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Task 6: Power BI Dashboard

Objective:
To create an interactive Power BI dashboard that presents key sales metrics and analysis in a clear and meaningful way.

Activities Performed:

1. Created KPI cards to display important sales metrics.
2. Created **Sales by Category** visualization.
3. Created **Sales Trend Over Time** visualization.
4. Created **Sales by Region** visualization.
5. Added a **Sales Details** table.
6. Added **Region** and **Category** slicers for filtering.
7. Added a meaningful dashboard title: **Sales Performance Dashboard**.
8. Organized the visuals and KPI cards into a clear dashboard layout.

Result:
An interactive **Sales Performance Dashboard** was created in Power BI, allowing users to view key sales metrics and analyze sales performance by category, product, region, and time period.

Task 7: Business Insights & Recommendations

Objective:
To interpret the analyzed sales data, identify key business insights, and provide data-driven recommendations.

Activities Performed:

1. Identified the highest-performing **category** based on sales.
2. Identified the highest-performing **region**.
3. Analyzed sales trends over time.
4. Identified high- and low-performing areas from the analysis.
5. Derived meaningful business insights from the Power BI dashboard.
6. Provided recommendations based on the identified sales patterns.

Key Insights:

- **Electronics** is the highest-performing category.
- **Kochi** is the highest-performing region.
- Sales performance varies across the available time periods.
- **Laptop** is among the high-performing products in the dataset.
- The highest recorded sales value is **₹70,000**, while the lowest is **₹2,500**.

Recommendations:

- Focus on high-performing Electronics products and maintain suitable inventory levels.
- Improve lower-performing categories through targeted promotions and offers.
- Use regional sales performance to plan marketing and inventory allocation.

Result:
The analysis provided actionable insights into sales performance and helped identify areas where the business can focus its sales and marketing efforts.

Task 8: Documentation & Project Presentation

Objective:
To document the complete project and present the analysis, findings, and recommendations in a clear and professional manner.

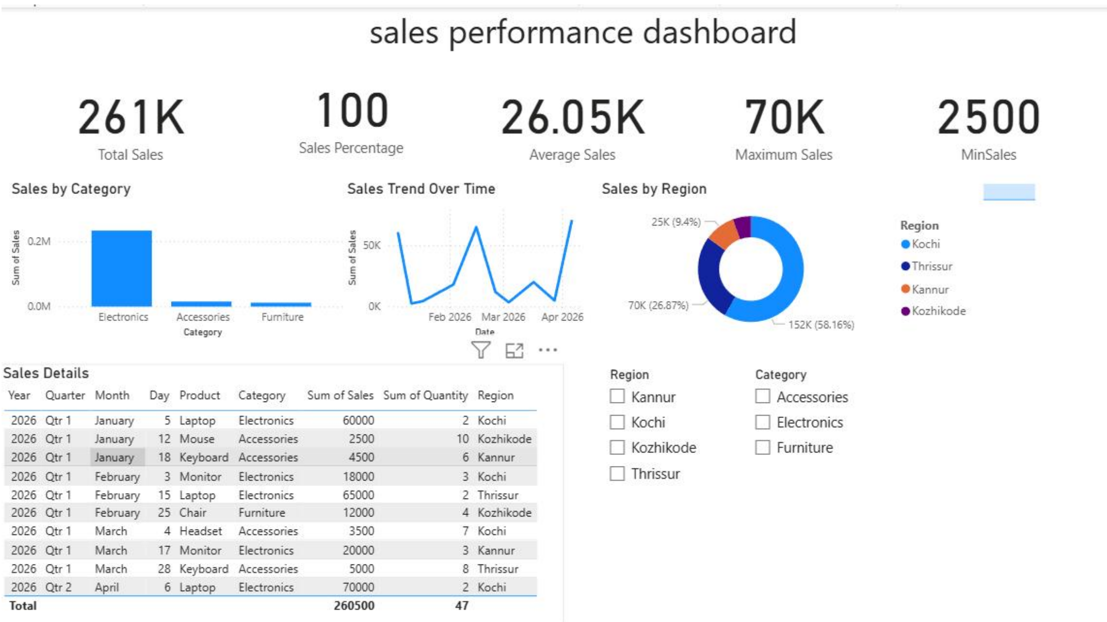
Activities Performed:

1. Prepared project documentation covering the major stages of the analysis.
2. Documented the **Project Overview** and **Problem Statement**.
3. Documented the **Dataset Description** and **Tools Used**.
4. Documented the **Data Cleaning & Preparation** process.
5. Documented the **Exploratory Data Analysis (EDA)**.
6. Documented the **Power BI Visualizations and Dashboard**.
7. Documented the **Key Insights and Business Recommendations**.

8. Prepared a short presentation structure covering the project objective, dataset, analysis, dashboard, findings, and conclusion.
9. Maintained the project files and documentation in the **GitHub repository**.

Result:

A complete project documentation and presentation structure was prepared to communicate the sales analysis, Power BI dashboard, key findings, and recommendations.



Year	Quarter	Month	Day	Product	Category	Sum of Sales	Sum of Quantity	Region
2026	Qtr 1	January	5	Laptop	Electronics	60000	2	Kochi
2026	Qtr 1	January	12	Mouse	Accessories	2500	10	Kozhikode
2026	Qtr 1	January	18	Keyboard	Accessories	4500	6	Kannur
2026	Qtr 1	February	3	Monitor	Electronics	18000	3	Kochi
2026	Qtr 1	February	15	Laptop	Electronics	65000	2	Thrissur
2026	Qtr 1	February	25	Chair	Furniture	12000	4	Kozhikode
2026	Qtr 1	March	4	Headset	Accessories	3500	7	Kochi
2026	Qtr 1	March	17	Monitor	Electronics	20000	3	Kannur
2026	Qtr 1	March	28	Keyboard	Accessories	5000	8	Thrissur
2026	Qtr 2	April	6	Laptop	Electronics	70000	2	Kochi
Total						260500	47	